RFC8950
Experience at BCIX

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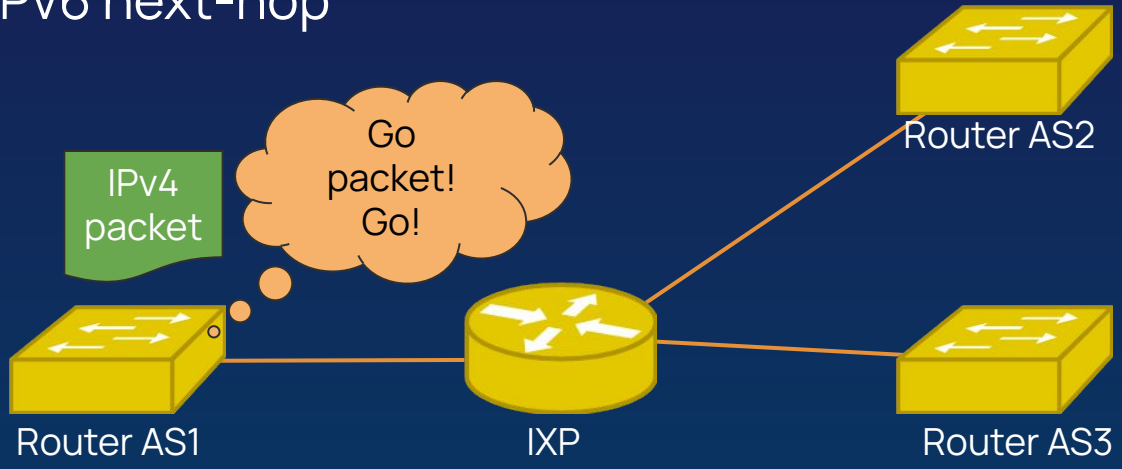
BCIX

What's RFC8950 (short version).

- Summary:
  - BGP capability Extended Next-hop; at session start
  - Next-hop IPv6 address in BGP NLRI → RIB → FIB
  - MAC lookup from IPv6 next-hop
- Just a few pieces!
- No tunneling!



Routers need to support this

- We tested a few (mostly virtual)
- Issues with old platforms
- Updates necessary
- Example Config:

  https://github.com/euro-ix/rfc8950-ixp

Vendor	OS	Status
Arista	EOS	Since 4.22.1F
Cisco	IOS XE	Not supported
Cisco	IOS XR	Since 7.3.3
Cisco	NX-OS	Supported, not tested
CZNIC	Bird (+ Linux)	Since Bird 2.0.8, Linux 5.2
Exa	ExaBGP	Since 4.1.0, only RIB
Extreme Networks	IronWare, SLX-OS	Not supported
FRR community	FRR (+ Linux)	FRR 9.1.3 (7.0.0), Linux 5.2
Huawei	VRP	Not fully supported
IPInfusion	OcNOS	Not fully supported
Juniper	JunOS	Since 21.2
Mikrotik	ROS	Since 7.20 (beta)
Nokia	SR-OS	Since 19.5R1
Nokia	SR Linux	Since 20.06, not tested
OpenBSD	OpenBGPd	8.8, only RIB
OSRG	GoBGP	Supported, only RIB
RtBrick	RBFS	All versions
Vyatta	VyOS	Since 1.4.3 (1.2.2)

Idea to enable RFC8950 on Route Servers

- We are using IXP Manager
  - Comes with templates for route server configuration
  - Multi-Table, separate daemon per AFI
- What we want
  - “Single” table – well, two tables (v4 + v6)
  - Two daemons:
    - IPv6 daemon with RFC8950 (master4 and master6)
    - IPv4 daemon as usual (master → master4)
  - Alice-LG

Multi-table vs. Single-table

- Traditionally RS used 1 table per peer to avoid route-hiding (caused by no-export communities)
- Bird allows the same with single sorted table
  - Best route is rejected by export filter → use next best
- Tests show: No compromise in route quality nor operations
- Single table reduces memory footprint and CPU overhead for pipes and filters

Options to enable RFC8950 on Route Servers

1. Use ARouteServer
   - We already implemented (and upstreamed) that
   - We can generate ARouteServer config from IXP Manager
2. Change IXP Manager templates
   - We can reuse existing logic, but have to re-arrange it
   - Filters between tables → import/export filters
   - We fixed a few issues (related to standard communities)

Results

- Single table
  - Memory footprint halved
  - IPv4 daemon: 1.4G RES, 2.0G VIRT
- IPv6 + RFC8950
  - Memory footprint similar to multi-table IPv4
  - IPv6 daemon: 2.5G RES, 3.7G VIRT
- 7 ASNs with RFC8950 enabled
  - +1 with IPv6 without extended next-hop (Ben)

Real-life issues

- Some peers may have IPv4 AFI enabled on IPv6 BGP session without Extended Next-hop capability
  - … and may try to send you a full-table → filtered
  - … and Bird will try to send routes
  - … and will flood your logs:

```
bird: pb_0359_as212232: Invalid NEXT_HOP attribute - mismatched
address family (2001:7f8:19:1::3ff6:fa for ipv4)
bird: pb_0359_as212232: Invalid route 103.161.207.0/24 withdrawn
```

So easy? Why did we wait to roll-out?

- We love Alice-LG (for multiple reasons)
- … but Birdwatcher only supported 1 AFI per BGP session
- Side note: It supports Bird single table
- So we “asked” Annika to implement Dual-AFI support
- This took some time, because lightwatcher

Looking Glass – lg.bcix.de

lg.bcix.de/routeservers/rs01-bcix-v6?q=BCIX

BCIX Route Server
Looking Glass

ROUTE SERVERS

rs01.bcix.de (IPv4)
bird 2.17.2

rs01.bcix.de (IPv6)
bird 2.17.2

rs02.bcix.de (IPv4)
bird 2.17.1

rs02.bcix.de (IPv6)
bird 2.17.1

rs01.bcix.de (IPv6)

BCIX

Go to: Established Down

BGP SESSIONS ESTABLISHED

Neighbour	ASN	State	Uptime	Description		Received	Accepted	Filtered	Exported
2001:7f8:19:1::f2f1:2	62193	up	a day	AS62193 - BCIX Management GmbH		19	17	2	68893
					IPv4	9	9		67
					IPv6	10	8	2	68826
2001:7f8:19:1::f2f1:1	62193	up	7 days	AS62193 - BCIX Management GmbH		11	8	3	80421
2001:7f8:19:1:0:3:43d5:1	213973	up	a day	AS213973 - BCIX Outreach		129	71	58	80431
					IPv4	56	55	1	18
					IPv6	73	16	57	80413

Looking Glass – lg.bcix.de

Filter by Network or Next-Hop

Go to: Accepted

ROUTES ACCEPTED

Showing **all** of **17** routes

Network	Next-Hop	AS Path	Local Pref.	MED	Origin
185.108.76.0/22	2001:7f8:19:1::7966:1	31078 206028	100	0	IGP
193.34.24.0/22	2001:7f8:19:1::7966:1	31078	100	0	IGP
195.160.172.0/24	2001:7f8:19:1::7966:1	31078 206028 49225	100	0	IGP
2001:678:4c0::/48	2001:7f8:19:1::7966:1	31078 208262 41948	100	0	IGP
2001:678:560::/48	2001:7f8:19:1::7966:1	31078 206028 49225	100	0	IGP
2001:678:598::/48	2001:7f8:19:1::7966:1	31078 31010	100	0	IGP
2001:67c:13e0::/48	2001:7f8:19:1::7966:1	31078 208262 198453	100	0	IGP
2001:67c:24ac::/48	2001:7f8:19:1::7966:1	31078 50575	100	0	IGP
217.115.0.0/20	2001:7f8:19:1::7966:1	31078	100	0	IGP
2a00:1328::/32	2001:7f8:19:1::7966:1	31078	100	0	IGP
2a0d:2940::/29	2001:7f8:19:1::7966:1	31078 206028	100	0	IGP
2a0f:c600::/29	2001:7f8:19:1::7966:1	31078 208262	100	0	IGP
31.220.136.0/21	2001:7f8:19:1::7966:1	31078	100	0	IGP

Last Reboot: **Thursday, August 21, 2025 5:50 AM**

Last Reconfig: **Thursday, September 18, 2025 2:47 PM**

Daemon is up and running

Generated **a few seconds ago**.
Next refresh **just now**.

RELATED NEIGHBORS

rs01.bcix.de (IPv4)
193.178.185.142 9 / 9 / 0 / 115046 up for 7 days

rs01.bcix.de (IPv6)
2001:7f8:19:1::7966:1 17 / 17 / 0 / 80480 up for 7 days

rs02.bcix.de (IPv4)
193.178.185.142 9 / 9 / 0 / 115046 up for 18 days

ADDRESS FAMILY

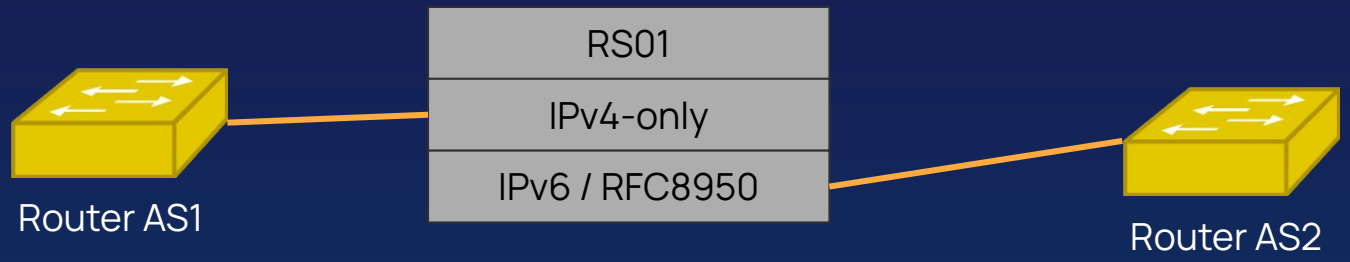
☑ IPv4 ☑ IPv6

BGP COMMUNITIES

Select BGP Communities to match...

After deploying RFC8950

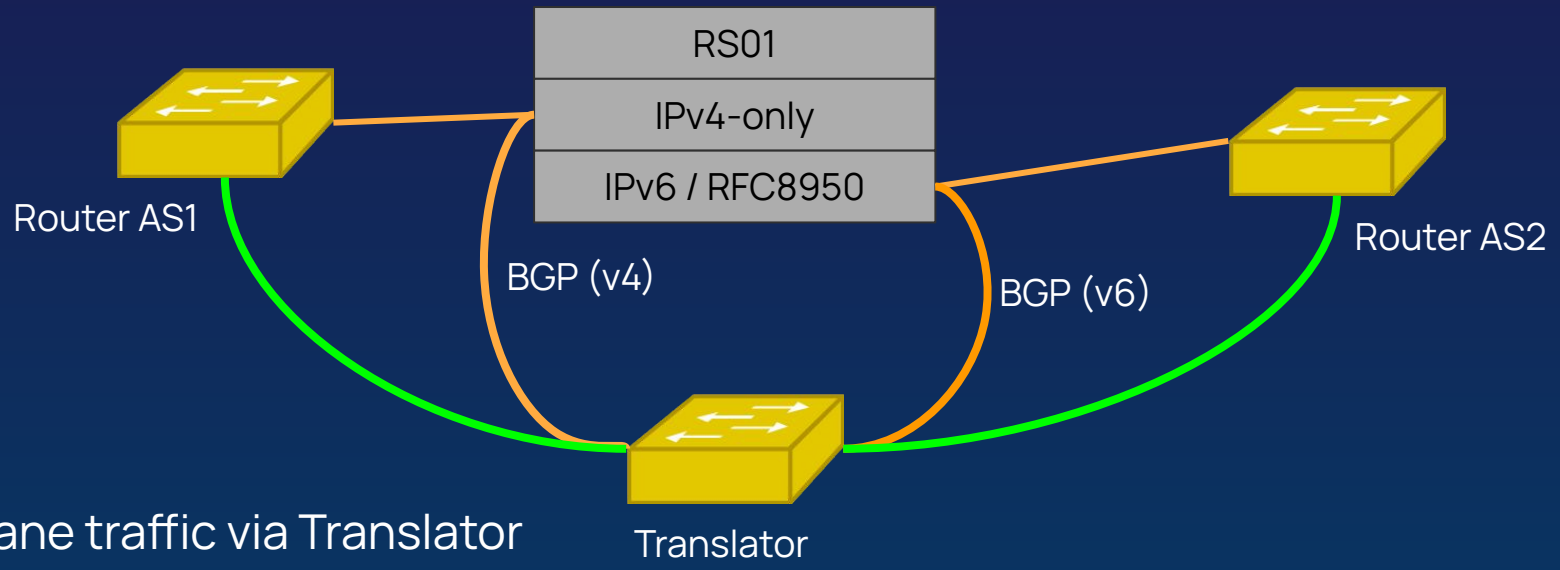
- There is IPv4 in IPv4-only and in RFC8950 part of RS



No dataplane traffic

Adding Translator

- Translator peers with RS “instances”, next-hop self



Dataplane traffic via Translator

The plan

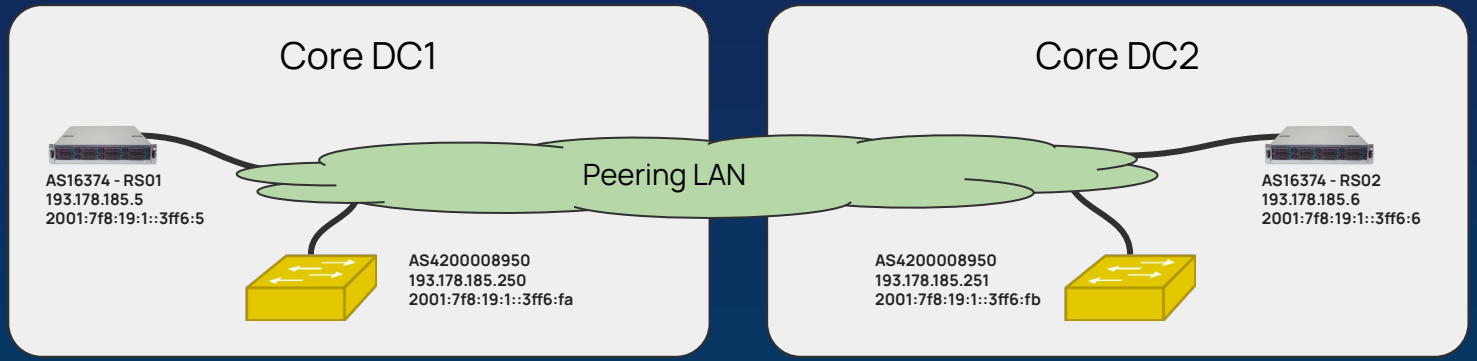
- 2 Route servers → we need 2 routers
- … connected with sufficient bandwidth
- Translator AS to be removed by route server
- Translated routes need lower priority than “native”
- No OTC added by RS towards translator (role: provider)
- Keep RS TE communities for handling on other end

Lower priority in BGP? Breaking ties!

Step	Comparison	Can we influence it?
1	Higher Local_pref	No
2	Shorter AS-Path	Yes, but negative effects on propagation
3	Origin (IGP, EGP, Incomplete)	Yes, but negative effects on propagation
4	Lowest MED	Yes, visible by route server client
5	eBGP before iBGP	No
6	IGP metric to next-hop	No
7	Oldest path	No
8	Lowest Neighbor router ID	Yes, towards Route-Server only

Our implementation (BCIX)

- We have 2x MX204 for the management network
- Located in same DC like route servers
- These have 2x 100G available → should be sufficient
- New routing instance, add IPv4+IPv6 addresses
- Configure BGP sessions to RS, no filters



Our implementation (BCIX)

- Create special peer config in route server
  - eBGP: AS4200008950
  - No filters (almost)
  - Ingress filter: remove first AS in Path
  - Egress filter: global do-not-export communities
- I.e. not an ordinary RS client in IXP Manager

RS implementation

```
filter f_export_as4200008950 {
    if (routeserverasn, 0, 0) ~ bgp_large_community then
        reject;
    if (0, routeserverasn) ~ bgp_community then
        reject;
    if bgp_large_community ~ [( routeserverasn, 1101, * )] then reject;

    #####Graceful BGP Session Shutdown####
    if (65535, 0) ~ bgp_community then {
        bgp_local_pref = 0;
    }
    accept;
}

filter f_import_as4200008950 {
    bgp_path.delete( 4200008950 );
    accept;
}
```

Avoid route hiding for no-export or invalid

Pretend the translator is not in the path.

Translator implementation (JunOS)

```
routing-instances {
  translator {
    instance-type virtual-router;
    routing-options {
      router-id 254.255.255.250;
      autonomous-system 4200008950;
      martians {
        240.0.0.0/4 orlonger allow;
      }
    }
    protocols {
      bgp {
        path-selection always-compare-med;
        group BCIX-v4 {
          type external;
          local-address 193.178.185.250;
          import translator-in;
          export [ translator-out then-reject ];
          peer-as 16374;
          neighbor 193.178.185.5;
        }
```

High router ID for worst prio.

Accept all

Next-hop self

```
        group BCIX-v6 {
          type external;
          local-address 2001:7f8:19:1:0::3ff6:fa;
          import translator-in;
          family inet {
            unicast {
              extended-nexthop;
            }
          }
          export [ translator-out then-reject ];
          peer-as 16374;
          neighbor 2001:7f8:19:1::3ff
        }
        metric-out 4200008950;
        bgp-error-tolerance {
          no-malformed-route-limit;
        }
      }
    }
  }
}
```

RFC8950

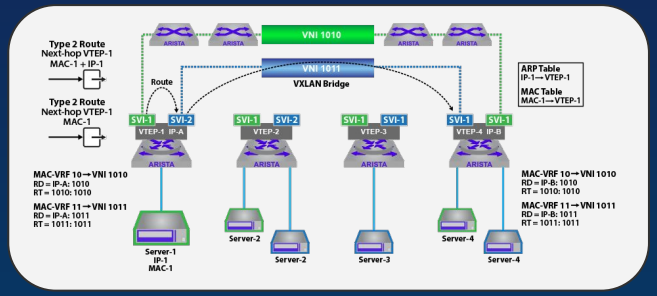
Worst MED

Caveats

- Only applies to routes available on RS
- Only one best path
  - “Don’t export” TE communities may cause route hiding
  - Add-Path not applicable
- We cannot scale bandwidth
  - Our peak so far: 20 Mbps

Scalability

- Idea: Integration into EVPN fabric
  - Employ Asymmetric IRB
  - Each edge switch uses virtual IPv4/IPv6 address on SVI
  - Challenge how to get routes from RS to all edge switches
    - IP reachability of edge switches via Peering LAN? Security?
    - Nokia SR-OS can leak BGP routes incl. next-hop
    - Arista cannot (yet?)

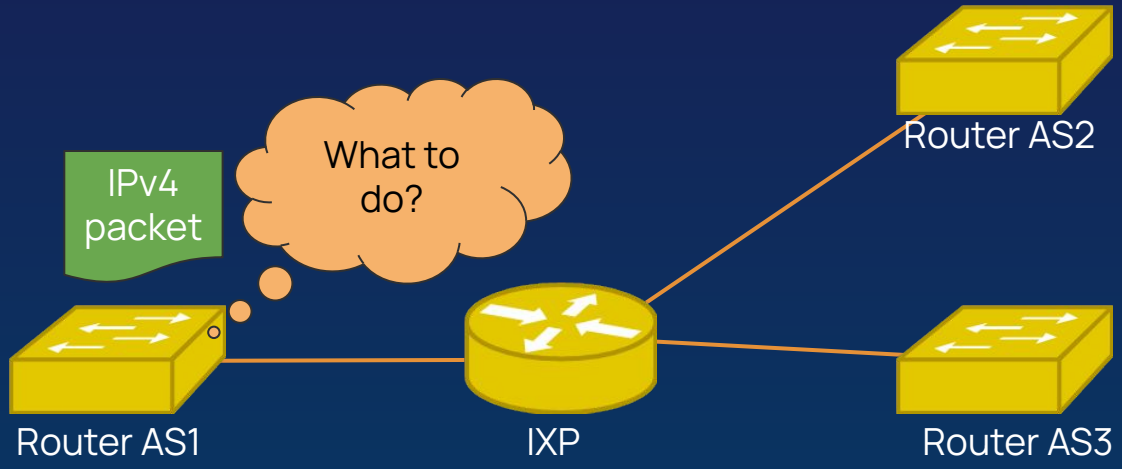


Spread the word!
Questions?
Ideas?

Backup slides

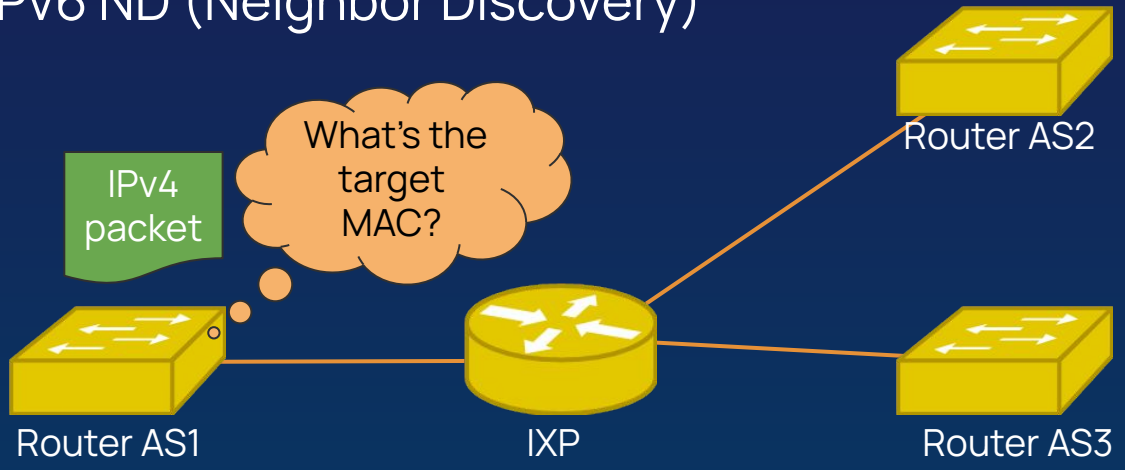
How does it work?

- Router's data plane receives packet
- Lookup destination IP in FIB, Longest-Prefix-Match (TCAM)
- Next-hop: target interface + target MAC



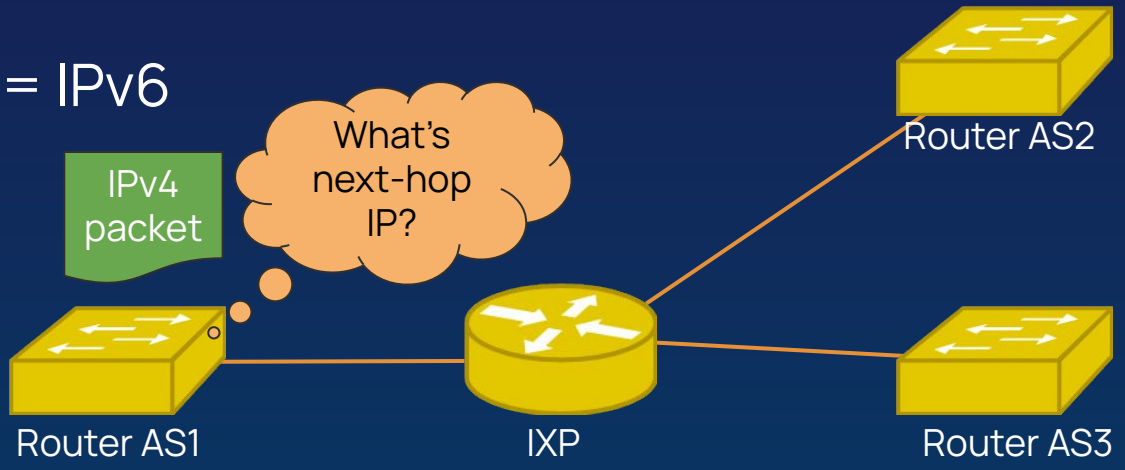
How does it work?

- To find target MAC a resolution may be necessary
- In RIB we have a next-hop IP address
  - Next-hop IPv4 → ARP
  - Next-hop IPv6 → IPv6 ND (Neighbor Discovery)



How does it work?

- Router learns next-hop IP from BGP
  - It's a variable length field
- RFC8950 re-defines interpretation:
  - 4 octets = IPv4
  - 16/32/48 octets = IPv6



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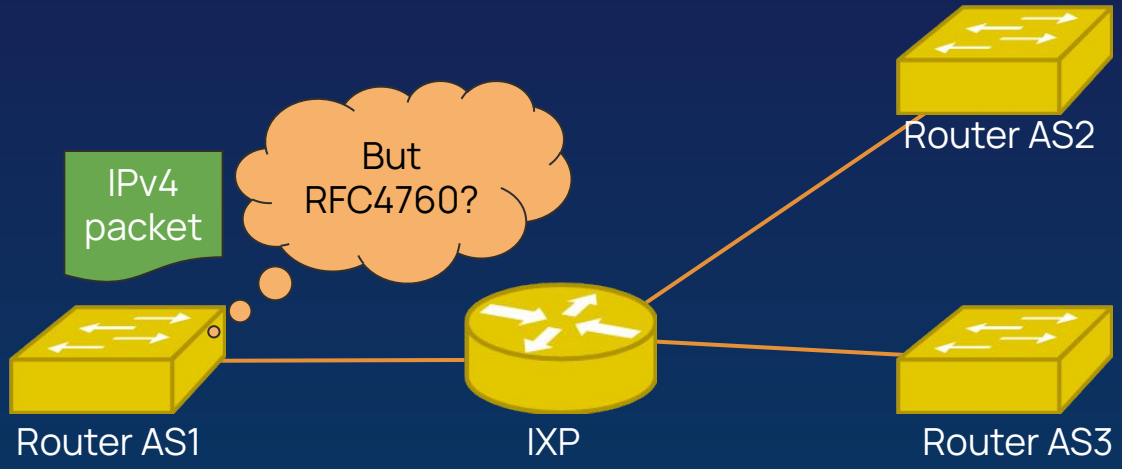
- Previously the address family of the next-hop was defined by the destination's AFI
- So we need a Capability advertised in BGP OPEN